

Workshop Replication

Name: Riya Patel

NUID: 002059816

Change Data Capture in Databricks

The screenshot shows the Databricks workspace interface. The top navigation bar includes the Databricks logo, a search bar, and a workspace dropdown. The left sidebar contains a 'New' button and a 'Workspace' section with various tools like SQL Editor, Queries, Dashboards, and more. The main area displays a SQL query in a notebook:

```
%sql
describe table extended workspace.damg7370.sample_employee_dataset_demo;
```

Below the query, a table of results is shown, detailing the table's metadata. The table has 25 rows and 1.07s runtime. The results are as follows:

col_name	data_type	comment
catalog	workspace	
Database	damg7370	
Table	sample_employee_dataset_demo	
Created Time	Sun Nov 02 19:53:27 UTC 2025	
Last Access	UNKNOWN	
Created By	Spark	
Statistics	1813 bytes, 6 rows	
Type	MANAGED	
Location		
Provider	delta	
Owner	patel.riyas@northeastern.edu	
Is_managed_location	true	
Predictive Optimization	ENABLE (inherited from METASTORE metastore_aws_us_east_2)	
Table Properties	{delta.enableDeletionVectors=true,delta.feature.appendOnly=supported,delta.feature.deletionVectors=supported,delta.feature.invariants=supported,delta.minReaderVersion=3,delta.minWriterVersion=7,delta.parquet.compression.codec=zstd}	

The bottom of the interface shows a Windows taskbar with various application icons, a search bar, and system information like temperature (45°F) and date (06-11-2025).

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Databricks x Raw_Data - Databricks x Untitled document - Google Docs x Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/314966830441506570=4201568283894456#command/4603608570539199

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

File Edit View Run Help Python Tabs: ON Last edit was 9 minutes ago

Run all Connected Schedule Share

07:08 PM (1s) 9

```
%sql
ALTER TABLE workspace.damg7370.sample_employee_dataset_demo
SET TBLPROPERTIES ('delta.enableChangeDataFeed' = true);
```

OK

07:08 PM (1s) 10

```
%sql
describe table extended workspace.damg7370.sample_employee_dataset_demo;
```

See performance (1) Optimize

Table	col_name	data_type	comment
1	Employee_ID	string	null
2	Name	string	null
3	Department	string	null
4	Performance_Score	bigint	null
5	Monthly_Salary	bigint	null
6			
7	# Delta Statistics Columns		
8	Column Names	Department, Name, Performance_Score, Monthly_Salary, Employee_ID	

Type here to search 45°F Clear 19:18 06-11-2025

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Databricks x Raw_Data - Databricks x Untitled document - Google Docs x Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/314966830441506570=4201568283894456#command/4603608570539199

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

File Edit View Run Help Python Tabs: ON Last edit was 9 minutes ago

Run all Connected Schedule Share

col_name	data_type	comment
10		
11	# Detailed Table Information	
12	Catalog	workspace
13	Database	damg7370
14	Table	sample_employee_dataset_demo
15	Created Time	Sun Nov 02 19:53:27 UTC 2025
16	Last Access	UNKNOWN
17	Created By	Spark
18	Statistics	1813 bytes, 6 rows
19	Type	MANAGED
20	Location	
21	Provider	delta
22	Owner	patel.riyas@northeastern.edu
23	Is_managed_location	true
24	Predictive Optimization	ENABLE (inherited from METASTORE metastore_aws_us_east_2)
25	Table Properties	{delta.enableChangeDataFeed=true,delta.enableDeletionVectors=true,delta.feature.appendOnly=supported,delta.feature.changeDataFeed=supported,delta.feature.deletionVectors=supported,delta.feature.invariants=supported,delta.minReaderVersion=3,delta.minWriterVersion=7,delta.parquet.compression.codec=zstd}

25 rows | 0.79s runtime Refreshed 10 minutes ago

This result is stored as _sqlidf and can be used in other Python and SQL cells.

Type here to search 45°F Clear 19:19 06-11-2025

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Databricks x Raw_Data - Databricks x Untitled document - Google Docs x Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/314966830441506570=4201568283894456#command/4603608570539199

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

File Edit View Run Help Python Tabs: ON ☆ Last edit was 9 minutes ago

Run all Connected Schedule Share

SQL

```
%sql
UPDATE workspace.dang7370.sample_employee_dataset_demo
SET Monthly_Salary = 7600
WHERE Employee_ID = 'E184';
```

See performance (1) Optimize

num_affected_rows
1

1 row | 4.47s runtime Refreshed 9 minutes ago

This result is stored as _sqlidf and can be used in other Python and SQL cells.

SQL

```
%sql
DESCRIBE HISTORY workspace.dang7370.sample_employee_dataset_demo;
```

See performance (1) Optimize

45°F Clear 19:19 06-11-2025

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Databricks x Raw_Data - Databricks x Untitled document - Google Docs x Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/314966830441506570=4201568283894456#command/4603608570539199

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

File Edit View Run Help Python Tabs: ON ☆ Last edit was 9 minutes ago

Run all Connected Schedule Share

SQL

```
%sql
DESCRIBE HISTORY workspace.dang7370.sample_employee_dataset_demo;
```

See performance (1) Optimize

version	timestamp	userid	userName	operation	operationParameters	
1	9	2025-11-07T00:09:29.000+00...	782945094599...	patel.riyas@northeastern.e...	OPTIMIZE	> ["clusterBy":[""],"zOrderBy":[""],"bab
2	8	2025-11-07T00:09:27.000+00...	782945094599...	patel.riyas@northeastern.e...	UPDATE	> ["predicate":["\"(Employee_ID#1258
3	7	2025-11-07T00:08:34.000+00...	782945094599...	patel.riyas@northeastern.e...	SET TBLPROPERTIES	> ["properties":["\"(delta.enableChang
4	6	2025-11-02T22:09:16.000+00...	782945094599...	patel.riyas@northeastern.e...	CREATE OR REPLACE TABLE AS SELECT	> ["partitionBy":[""],"clusterBy":[""],"d
5	5	2025-11-02T21:16:29.000+00...	782945094599...	patel.riyas@northeastern.e...	WRITE	> ["mode":["Append"],"statsOnLoad":["
6	4	2025-11-02T21:12:26.000+00...	782945094599...	patel.riyas@northeastern.e...	DELETE	> ["predicate":["\"(Employee_ID#1148
7	3	2025-11-02T21:07:25.000+00...	782945094599...	patel.riyas@northeastern.e...	WRITE	> ["mode":["Append"],"statsOnLoad":["
8	2	2025-11-02T19:58:05.000+00...	782945094599...	patel.riyas@northeastern.e...	OPTIMIZE	> ["clusterBy":[""],"zOrderBy":[""],"bab
9	1	2025-11-02T19:58:02.000+00...	782945094599...	patel.riyas@northeastern.e...	DELETE	> ["predicate":["\"(Employee_ID#1177
10						

10 rows | 1.57s runtime Refreshed 10 minutes ago

This result is stored as _sqlidf and can be used in other Python and SQL cells.

45°F Clear 19:19 06-11-2025

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Databricks x Raw_Data - Databricks x Untitled document - Google Docs x Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/314966830441506570=4201568283894456&command/4603608570539199

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

File Edit View Run Help Python Tabs: ON

Last edit was 10 minutes ago

Run all Connected Schedule Share

SQL

```
%sql
select * from workspace.damg7370.sample_employee_dataset_deno
> See performance (1)
```

_sqlidf: pyspark.sql.connect.dataframe.DataFrame = [Employee_ID: string, Name: string ... 3 more fields]

Table	Employee_ID	Name	Department	Performance_Score	Monthly_Salary
1	E104	Diana Chen	Human Resources	85	7600
2	E101	Alice Johnson	Marketing	88	5400
3	E102	Bob Smith	Finance	92	6000
4	E103	Charlie Patel	Engineering	76	6800
5	E105	Eric Lopez	Engineering	90	7100
6	E106	Fiona Davis	Marketing	79	5000

6 rows | 1.53s runtime Refreshed 10 minutes ago

This result is stored as _sqlidf and can be used in other Python and SQL cells.

[Shift+Enter] to run and move to next cell
[Ctrl+Shift+P] to open the command palette
[Esc H] to see all keyboard shortcuts

Type here to search 45°F Clear 19:19 06-11-2025

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Databricks x PL_DLT - Databricks x Untitled document - Google Docs x Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/314966830441506470=4201568283894456&contextId=pipeline%3Ae6e84399-c13c-49b1-aad0-eed1d7bb1c81&command...

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

PL_DLT

Table of contents

A table of contents will appear here when you add cell titles or Markdown headings.

Markdown heading syntax:
<h1># Heading 1</h1>

Medallion_DLT

Python Lakeflow Pipelines Editor: ON

```
1 import dlt
2 from pyspark.sql.functions import *
3 from pyspark.sql.types import *
```

Pipeline graph

Maximized

Tables 5 Performance 4

Name	Catalog	Schema	Type	Dur...	Out...	Expectati...	Dro...	Wa...	Fail...	Incremen...
silver_em...	N/A	N/A	View	-	-	Not defined	-	-	-	N/A
bronze_e...	workspace	damg7370	Materializ...	3s	6	Not defined	-	-	-	Full reco
gold_em...	workspace	damg7370	Materializ...	6s	6	Not defined	-	-	-	Incremei
sample_t...	workspace	damg7370	Materializ...	5s	22K	Not defined	-	-	-	No chan
sample_z...	workspace	damg7370	Materializ...	2s	128	Not defined	-	-	-	No chan

Nov 06, 2025, 07:10 PM 36s Refresh all Query performance

Type here to search 45°F Clear 19:20 06-11-2025

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Data: x bronze_employee - Databricks x Untitled document - Google D: x Gemini x

dbc-6043e27f-1bee.cloud.databricks.com/explore/data/workspace/damg7370/bronze_employee?o=4201568283894456&activeTab=sample

databricks Search data, notebooks, recents, and more... CTRL + P workspace

new

- Workspace
- Recents
- Catalog**
- Jobs & Pipelines
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion

AI/ML

- Playground
- Experiments

Catalog

- Serverless Starter Warehouse Serverless 2XS
- Type to search...
- My organization
 - workspace
 - damg7370
 - Tables (8)
 - bronze_employee**
 - gold_employee_demo
 - ny_health_data
 - sample_employee_dataset
 - sample_employee_dataset_demo
 - sample_trips_pl_dlt
 - sample_zones_pl_dlt
 - test
 - Volumes (1)
 - default
 - information_schema
 - system
 - Delta Shares Received
 - samples

bronze_employee Open in a dashboard Share Create

Overview **Sample Data** Details Permissions Policies History Lineage Insights Quality

Ask your question about the sample data... Preview Reset

Are there any departments with below-average performance scores? Which employees have the highest monthly salaries? Are there any

Sample

	Employee_ID	Name	Department	Performance_Score	Monthly_Salary
1	E104	Diana Chen	Human Resources	85	7600
2	E101	Alice Johnson	Marketing	88	5400
3	E102	Bob Smith	Finance	92	6000
4	E103	Charlie Patel	Engineering	76	6800
5	E105	Eric Lopez	Engineering	90	7100
6	E106	Fiona Davis	Marketing	79	5000

Change Data Capture CDC x (17) Feed | LinkedIn x Free Edition | Replacing Data: x gold_employee_demo - Databricks x Untitled document - Google D: x Gemini x

dbc-6043e27f-1bee.cloud.databricks.com/explore/data/workspace/damg7370/gold_employee_demo?o=4201568283894456&activeTab=sample

databricks Search data, notebooks, recents, and more... CTRL + P workspace

new

- Workspace
- Recents
- Catalog**
- Jobs & Pipelines
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion

AI/ML

- Playground
- Experiments

Catalog

- Serverless Starter Warehouse Serverless 2XS
- Type to search...
- My organization
 - workspace
 - damg7370
 - Tables (8)
 - bronze_employee
 - gold_employee_demo**
 - ny_health_data
 - sample_employee_dataset
 - sample_employee_dataset_demo
 - sample_trips_pl_dlt
 - sample_zones_pl_dlt
 - test
 - Volumes (1)
 - default
 - information_schema
 - system
 - Delta Shares Received
 - samples

gold_employee_demo Open in a dashboard Share Create

Overview **Sample Data** Details Permissions Policies History Lineage Insights Quality

Ask your question about the sample data... Preview Reset

Average salary by department? Top 5 performers by score? Any inactive employee records?

Sample

	Employee_ID	Name	Department	Performance_Score	Monthly_Salary	row_status
1	E104	Diana Chen	Human Resources	85	7600.0	New
2	E101	Alice Johnson	Marketing	88	5400.0	New
3	E102	Bob Smith	Finance	92	6000.0	New
4	E103	Charlie Patel	Engineering	76	6800.0	New
5	E105	Eric Lopez	Engineering	90	7100.0	New
6	E106	Fiona Davis	Marketing	79	5000.0	New

SCD Type 1 and 2 in Databricks

The screenshot displays the Databricks Jobs & Pipelines interface for the pipeline **PL_CDF_SCD1_SCD2_EMP**. The pipeline is in a "Completed" state, having run on Nov 07, 2025, at 02:56 PM. The interface shows a graph of the pipeline's stages: **bronze_employee_cdf**, **silver_employee_cdf**, **gold_employee_cdf**, **employee_cdf_type2**, **sample_users_pl_d...**, and **sample_aggregati...**. The right-hand panel provides details about the pipeline, including its ID (9c9a61ea-4f9e-433f-baf6-7697780796dd), type (ETL pipeline), and source code. Below the graph, the "Event log" shows a series of completion messages for the pipeline's flows and updates.

Pipeline details

- Pipeline ID: 9c9a61ea-4f9e-433f-baf6-7697780796dd
- Pipeline type: ETL pipeline
- Source code: /Workspace/Users/patel.riyas@northeastern.edu/PL_CDF_SCD1_SCD2_EMP/transformatio ns/

Event log

Time	Event Type	Message
21 hours ago	flow_progress	Flow 'workspace.damg7370.sample_aggregation_pl_cdf_scd1_scd2_emp' has COMPLETED.
21 hours ago	flow_progress	Completed a streaming update of 'workspace.damg7370.employee_cdf_type2_stage'.
21 hours ago	flow_progress	Flow 'workspace.damg7370.employee_cdf_type2_stage' has COMPLETED.
21 hours ago	update_progress	Update 520eb6 is COMPLETED.

The screenshot displays the Databricks Jobs & Pipelines interface for the pipeline **PL_DLT_SCD2_Slv_To_Gld**. The pipeline is in a "Completed" state, having run on Nov 08, 2025, at 11:43 AM. The interface shows a graph of the pipeline's stages: **silver_employee_cdf**, **gold_employee_cdf**, **sample_users_pl_d...**, and **sample_aggregati...**. The right-hand panel provides details about the pipeline, including its ID (9cf36e23-9b92-43af-a402-7d460cd05060), type (ETL pipeline), and source code. Below the graph, the "Event log" shows a series of completion messages for the pipeline's dataset definitions and updates.

Pipeline details

- Pipeline ID: 9cf36e23-9b92-43af-a402-7d460cd05060
- Pipeline type: ETL pipeline
- Source code: /Workspace/Users/patel.riyas@northeastern.edu/PL_DLT_SCD2_Slv_To_Gld/transformatio ns/

Event log

Time	Event Type	Message
13 minutes ago	dataset_definition	Dataset 'workspace.'damg7370'.sample_aggregation_pl_dlt_scd2_slv_to_gld defined as MATERIALIZED_VIEW.
13 minutes ago	dataset_definition	Dataset 'silver_employee_cdf_type2_stage' defined as VIEW.
13 minutes ago	dataset_definition	Dataset 'workspace.'damg7370'.gold_employee_cdf_streaming_type2 defined as STREAMING_TABLE.
13 minutes ago	update_progress	Update c79ae7 is COMPLETED.

satge_2_gold_scd2 - Databricks x Catalog Explorer - Databricks x Riya - medallion x Workshop Replication_Change x + Gemini - + x

dbc-6043e27f-1bee.cloud.databricks.com/explore/data?o=4201568283894456

Search data, notebooks, recents, and more... CTRL + P workspace

New

- Workspace
- Recents
- Catalog**
- Jobs & Pipelines
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion

AI/ML

- Playground
- Experiments

Catalog

Type to search...

- My organization
 - workspace
 - damg7370
 - Tables (14)
 - bronze_employee
 - bronze_employee_cdf
 - employee_cdf_type2_stage
 - gold_employee_cdf_streaming_type1
 - gold_employee_demo
 - ny_health_data
 - sample_aggregation_pl_cdf_scd1_scd2_emp
 - sample_employee_cdf
 - sample_employee_dataset
 - sample_employee_dataset_demo
 - sample_trips_pl_dlt
 - sample_users_pl_cdf_scd1_scd2_emp
 - sample_zones_pl_dlt
 - test
 - Volumes (1)
 - default
 - information_schema

Quick access

Recents Favorites Catalogs Filter

Name	Last viewed	Type
damg7370 workspace	50 minutes ago	Schema
workspace	50 minutes ago	Catalog
bronze_employee_cdf workspace.damg7370	19 hours ago	Streaming table
gold_employee_cdf_streaming_type1 workspace.damg7370	21 hours ago	Streaming table
default workspace	21 hours ago	Schema
gold_employee_demo workspace.damg7370	1 day ago	Materialized view
bronze_employee workspace.damg7370	1 day ago	Materialized view
sample_employee_dataset_demo workspace.damg7370	5 days ago	Table
sample_employee_dataset workspace.damg7370	5 days ago	Table
ny_health_data		

Type here to search

1 inch of rain Mon

ENG 11:55 08-11-2025

satge_2_gold_scd2 - Databricks x bronze_employee_cdf - Databricks x Riya - medallion x Workshop Replication_Change x + Gemini - + x

dbc-6043e27f-1bee.cloud.databricks.com/explore/data/workspace/damg7370/bronze_employee_cdf?o=4201568283894456&activeTab=sample

Search data, notebooks, recents, and more... CTRL + P

New

- Workspace
- Recents
- Catalog**
- Jobs & Pipelines
- Compute
- Marketplace

SQL

- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses

Data Engineering

- Job Runs
- Data Ingestion

AI/ML

- Playground
- Experiments

Catalog

Type to search...

- My organization
 - workspace
 - damg7370
 - Tables (14)
 - bronze_employee
 - bronze_employee_cdf**
 - employee_cdf_type2_stage
 - gold_employee_cdf_streaming_type1
 - gold_employee_demo
 - ny_health_data
 - sample_aggregation_pl_cdf_scd1_scd2_emp
 - sample_employee_cdf
 - sample_employee_dataset
 - sample_employee_dataset_demo
 - sample_trips_pl_dlt
 - sample_users_pl_cdf_scd1_scd2_emp
 - sample_zones_pl_dlt
 - test
 - Volumes (1)
 - default
 - information_schema

Serverless Starter Warehouse has started.
See compute details for more information.

Catalog Explorer > workspace > damg7370 >

bronze_employee_cdf

Overview **Sample Data** Details Permissions Policies History Lineage Insights Quality

Ask your question about the sample data... Preview Reset

Count new employees by department per month. Average monthly salary per performance score. Top 5 departments with highest salary

Sample

	Employee_ID	Name	Department	Performance_Score	Monthly_Salary	change_type
1	E104	Diana Chen	Human Resources	85	7600	insert
2	E101	Alice Johnson	Marketing	88	5400	insert
3	E102	Bob Smith	Finance	92	6000	insert
4	E103	Charlie Patel	Engineering	76	6800	insert
5	E105	Eric Lopez	Engineering	90	7100	insert
6	E106	Fiona Davis	Marketing	79	5000	insert
7	E104	Diana Chen	Human Resources	85	7600	update_preimage
8	E104	Diana Chen	Human Resources	85	100000	update_postimage

Type here to search

Temps to plummet

ENG 11:57 08-11-2025

satge_2_gold_scd2 - Databricks x gold_employee_cdf_streaming... x Riya - medallion x Workshop Replication_Change x + Gemini

dbc-6043e27f-1bee.cloud.databricks.com/explore/data/workspace/damg7370/gold_employee_cdf_streaming_type1?o=4201568283894456&activeTab=sample

Search data, notebooks, recents, and more... CTRL + P workspace

New

- Workspace
- Recents
- Catalog**
- Jobs & Pipelines
- Compute
- Marketplace
- SQL
- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses
- Data Engineering
- Job Runs
- Data Ingestion
- AI/ML
- Playground
- Experiments

Catalog

- Serverless Starter Warehouse Serverless 2XS
- Type to search...
- My organization
 - workspace
 - damg7370
 - Tables (14)
 - bronze_employee
 - bronze_employee_cdf
 - employee_cdf_type2_stage
 - gold_employee_cdf_streaming_type1**
 - gold_employee_demo
 - ny_health_data
 - sample_aggregation_pl_cdf_scd1_scd2_emp
 - sample_employee_cdf
 - sample_employee_dataset
 - sample_employee_dataset_demo
 - sample_trips_pl_dlt
 - sample_users_pl_cdf_scd1_scd2_emp
 - sample_zones_pl_dlt
 - test
 - Volumes (1)
 - default
 - information_schema

Catalog Explorer > workspace > damg7370 >

gold_employee_cdf_streaming_type1 ☆

Open in a dashboard ▾ Share ▾ Create ▾

Overview **Sample Data** Details Permissions Policies History Lineage Insights Quality

Ask your question about the sample data... Preview ▶ Reset

What is the average Monthly_Salary by Department? How many employees have a Performance_Score of 5? Are there any salary outliers

Sample

	Employee_ID	Name	Department	Performance_Score	Monthly_Salary	change_type
1	E101	Alice Johnson	Marketing	88	5400.0	insert
2	E102	Bob Smith	Finance	92	6000.0	insert
3	E103	Charlie Patel	Engineering	76	6800.0	insert
4	E105	Eric Lopez	Engineering	90	7100.0	insert
5	E106	Fiona Davis	Marketing	79	5000.0	insert
6	E104	Diana Chen	Human Resources	85	10000.0	update_postimage

satge_2_gold_scd2 - Databricks x employee_cdf_type2_stage - D... x Riya - medallion x Workshop Replication_Change x + Gemini

dbc-6043e27f-1bee.cloud.databricks.com/explore/data/workspace/damg7370/employee_cdf_type2_stage?o=4201568283894456&activeTab=sample

Search data, notebooks, recents, and more... CTRL + P workspace

New

- Workspace
- Recents
- Catalog**
- Jobs & Pipelines
- Compute
- Marketplace
- SQL
- SQL Editor
- Queries
- Dashboards
- Genie
- Alerts
- Query History
- SQL Warehouses
- Data Engineering
- Job Runs
- Data Ingestion
- AI/ML
- Playground
- Experiments

Catalog

- Serverless Starter Warehouse Serverless 2XS
- Type to search...
- My organization
 - workspace
 - damg7370
 - Tables (14)
 - bronze_employee
 - bronze_employee_cdf
 - employee_cdf_type2_stage**
 - gold_employee_cdf_streaming_type1
 - gold_employee_demo
 - ny_health_data
 - sample_aggregation_pl_cdf_scd1_scd2_emp
 - sample_employee_cdf
 - sample_employee_dataset
 - sample_employee_dataset_demo
 - sample_trips_pl_dlt
 - sample_users_pl_cdf_scd1_scd2_emp
 - sample_zones_pl_dlt
 - test
 - Volumes (1)
 - default
 - information_schema

Catalog Explorer > workspace > damg7370 >

employee_cdf_type2_stage ☆

Open in a dashboard ▾ Share ▾ Create ▾

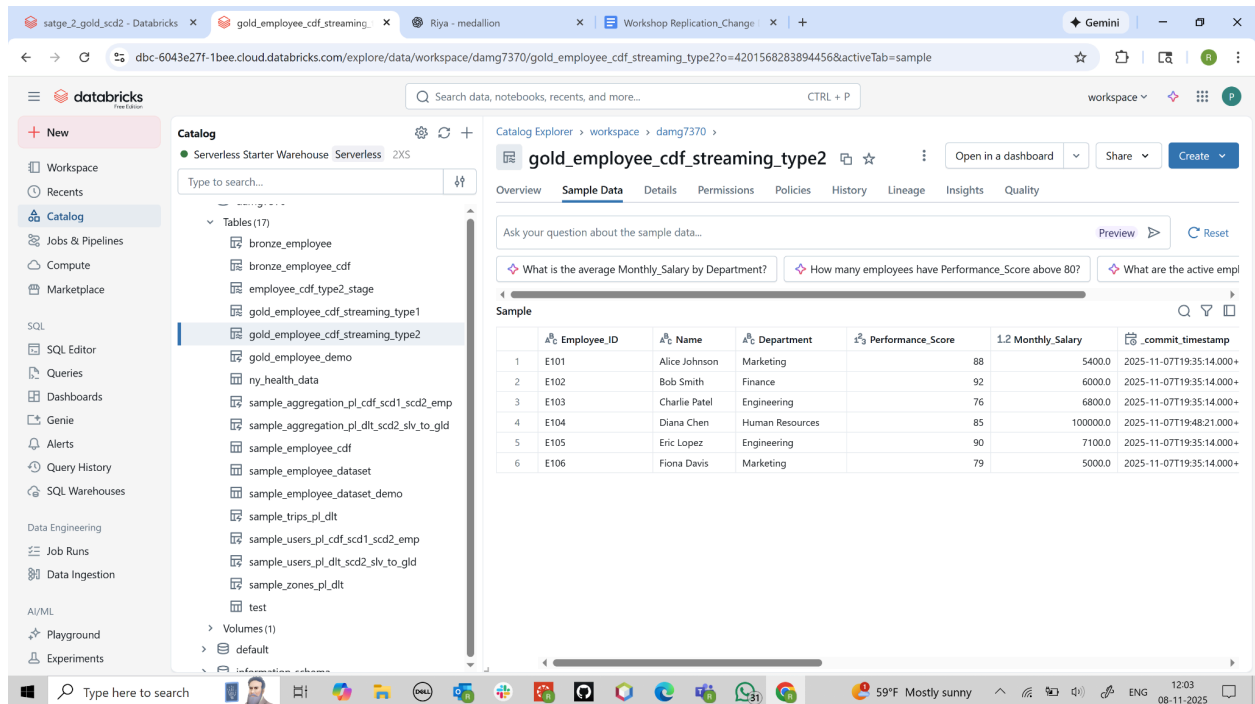
Overview **Sample Data** Details Permissions Policies History Lineage Insights Quality

Ask your question about the sample data... Preview ▶ Reset

What is the average performance score by department? Which employees have the highest monthly salary? Are there any performance s

Sample

	Employee_ID	Name	Department	Performance_Score	Monthly_Salary	change_type
1	E101	Alice Johnson	Marketing	88	5400.0	insert
2	E102	Bob Smith	Finance	92	6000.0	insert
3	E103	Charlie Patel	Engineering	76	6800.0	insert
4	E104	Diana Chen	Human Resources	85	7600.0	insert
5	E105	Diana Chen	Human Resources	85	10000.0	update_postimage
6	E105	Eric Lopez	Engineering	90	7100.0	insert
7	E106	Fiona Davis	Marketing	79	5000.0	insert



Error in Code

Identify if there is any bug in the code

Yes, there was a bug in the original code. The `read.table()` command performs a *static read* and does not support streaming, which caused the pipeline initialization to fail when Databricks tried to process the data dynamically. Since the Gold layer depends on the Silver layer as a streaming input, this mismatch prevented the pipeline from starting correctly.

Will the pipeline execute without any errors? Explain

No, the pipeline did not execute successfully at first because streaming data cannot be read using `read.table()`. Delta Live Tables (DLT) pipelines are designed to work with streaming dataframes for continuous updates, and using a static read breaks that flow. Once the code was corrected to use the pipeline ran successfully. The `readStream` function allows the Silver layer to continuously process changes from the Bronze or CDC table in real time, which fixed the initialization failure and enabled the Gold layer to apply CDC logic smoothly.

If the pipeline fails, correct the code so the implementation succeeds

To correct the error, the code was modified to use the streaming read method.

The corrected Silver layer code is:

`@dlt.view`

`def silver_employee_cdf_type2_stage():`

`df = spark.readStream.table("employee_cdf_type2_stage")`

`df = df.withColumn("start_dt", col("__START_AT"))`

`df = df.withColumn("end_dt", col("__END_AT"))`

`df = df.withColumn(`

```

    "is_active",
    when(col("end_dt").isNull(), lit("Y")).otherwise(lit("N"))
)
df = df.withColumn("load_dt", current_timestamp())
return df.drop("__START_AT", "__END_AT")

```

After this change, the pipeline was able to initialize properly and run end-to-end without errors.

Final Approach

The approach followed the Medallion Architecture pattern (Bronze → Silver → Gold).

- The Bronze layer captured the CDC-enabled raw data stream from the Delta table.
- The Silver layer (where the error occurred) was meant to clean and enrich the streamed data. The fix ensured it continuously listens to CDC changes instead of performing a one-time read.
- The Gold layer applied the CDC merge logic with Slowly Changing Dimension (SCD) tracking, storing the historical changes for each employee record.

By switching from `read.table()` to `readStream.table()`, the Silver layer could now process streaming updates from the Bronze table in real time, aligning perfectly with the CDC pipeline requirements.

The screenshot displays the Databricks interface with the following elements:

- Left Sidebar:** Navigation menu including Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Marketplace, SQL (SQL Editor, Queries, Dashboards, Genie), Alerts, Query History, SQL Warehouses, Data Engineering (Job Runs, Data Ingestion), and AI/ML (Playground, Experiments).
- Top Bar:** Search bar, workspace name 'satge_2_gold_scd2', and various icons for workspace management.
- Central Panel:**
 - Pipeline Configuration:** Shows the pipeline 'PL_DLT_SCD2_Slv_To_Gld' with a last run of 13s. Assets include explorations, transformations, and utilities.
 - Code Editor:** Displays a Python script using `delta.apply_changes` to merge data from a streaming source into a target table. The script includes parameters for target, source, keys, sequence_by, ignore_null_updates, apply_as_deletes, except_column_list, and stored_as_scd_type.
- Right Panel:**
 - Notification:** 'Pipeline settings updated' with a link to view details.
 - Error Message:** 'Graph unavailable' and 'Pipeline initialization failed: An error occurred during pipeline initialization'.
- Bottom Bar:** Windows taskbar showing the time as 11:41 on 08-11-2025.

satge_2_gold_scd2 - Databricks x PL_DLT_SCD2_Slv_To_Gld - Databricks x Krishna-medallion x Workshop Replication_Change x + Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/4046855463072989?contextid=pipeline%3A9cf36e23-9b92-43af-a402-7d460cd05060&o=4201568283894456#command/...

Pipeline settings updated
File "satge_2_gold_scd2" has been included in your pipeline source code configuration

PL_DLT_SCD2_Slv_To_Gld

Pipeline configuration: All files x

Last runs: 13s

Pipeline assets: + Add

- explorations
- transformations
 - sample_aggregation_pl_dlt_scd2...
 - sample_users_pl_dlt_scd2_slv_to...
- utilities
 - utils.py
 - README.md
 - satge_2_gold_scd2

```
dlt.create_streaming_table("gold_employee_cdf_streaming_type2")
dlt.apply_changes(
    target = "gold_employee_cdf_streaming_type2",
    source = "silver_employee_cdf_type2_stage",
    keys = ["Employee_ID"],
    sequence_by = struct("Employee_ID", "commit_timestamp"),
    ignore_null_updates = True, # columns with a null retain existing values in the target during update
    apply_as_deletes = expr("_change_type = 'delete'"),
    except_column_list = ["_change_type", "_commit_version"],
    stored_as_scd_type = 1
)
```

Tables 4 Performance

Name	Catalog	Schema	Type	Dur...	Out...	Ups...	Expectati...	Dro...	Wa...	Fail...	Del...
silver_e...	N/A	N/A	View	-	-	-	Not defin...	-	-	-	-
gold_em...	workspace	dmg7370	Streamin...	-	-	-	Not defin...	-	-	-	-
sample_...	workspace	dmg7370	Materializ...	-	-	-	Not defin...	-	-	-	-
sample_...	workspace	dmg7370	Materializ...	-	-	-	Not defin...	-	-	-	-

Nov 08, 2025, 11:43 AM · 13s · Dry run · Query performance

satge_2_gold_scd2 - Databricks x CDF - Databricks x Riya - medallion x Workshop Replication_Change x + Gemini

dbc-6043e27f-1bee.cloud.databricks.com/editor/notebooks/1437629291276627?o=4201568283894456#command/4609994261415286

Workspace

- PL_CDF_SCD1_SCD2_EMP
- PL_DLT
- PL_DLT_SCD2_Slv_To_Gld
- Bakehouse Sales Starter Space
- CDF
- Medallion_DLT
- Raw_Data
- Riya Patel_Basic Commands
- Riya Patel_Basics_Spark_Dataframe
- Workspace Usage Dashboard

CDF

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

```
%sql
select * from workspace.dmg7370.gold_employee_cdf_streaming_type2
```

See performance (1)

`_sqldf: pyspark.sql.connect.dataframe.DataFrame = [Employee_ID: string, Name: string ... 9 more fields]`

Employee_ID	Name	Department	Performance_Score	Monthly_Salary	commit_timest	
1	E101	Alice Johnson	Marketing	88	5400	2025-11-07T19:35:1
2	E102	Bob Smith	Finance	92	6000	2025-11-07T19:35:1
3	E103	Charlie Patel	Engineering	76	6800	2025-11-07T19:35:1
4	E104	Diana Chen	Human Resources	85	100000	2025-11-07T19:48:2
5	E105	Eric Lopez	Engineering	90	7100	2025-11-07T19:35:1
6	E106	Fiona Garcia	Marketing	82	5800	2025-11-07T19:35:1

6 rows | 32.81s runtime

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

PL_JOB

Run 1062371642397144 of New Job Nov 08, 2025, 02:47 PM

Code modification review

Workshop Replication_CDC an...

Untitled document - Google D...

Gemini

dbc-6043e27f-1bee.cloud.databricks.com/jobs/1079862481251979/runs/1062371642397144?o=4201568283894456

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

Jobs & Pipelines > New Job Nov 08, 2025, 02:47 PM >

New Job Nov 08, 2025, 02:47 PM run

Lakeflow Jobs UI: ON

Send feedback

Delete job run

Repair run

Graph

Timeline

List

Raw_to_Bronze

Succeeded · 1m 31s

Pipeline: PL_CDF_SCD1_SCD2_EMP

Silver_to_Gold

Succeeded · 1m 21s

Pipeline: PL_DLT_SCD2_Slv_To_Gld

Job run details

Job ID1079862481251979

Job run ID1062371642397144

LaunchedManually

StartedNov 08, 2025, 02:48 PM

EndedNov 08, 2025, 02:51 PM

Duration2m 54s

Execution time2m 53s

Queue duration-

StatusSucceeded

LineageNo lineage information for this job.
Learn more

Performance optimizationEnabled

View run events

Type here to search

62°F Sunny

14:52

08-11-2025

PL_JOB - Databricks

Code modification review

Workshop Replication_CDC an: X

Untitled document - Google D: X

dbc-6043e27f-1bee.cloud.databricks.com/jobs/1079862481251979/tasks/Raw_to_Bronze?o=4201568283894456

Search data, notebooks, recents, and more... CTRL + P

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

Jobs & Pipelines >

PL_JOB ☆ Lakeflow Jobs UI: ON Send feedback

Runs Tasks

Raw_to_Bronze

Pipeline: PL_CDF_SCD1_SCD2_EMP

Silver_to_Gold

Pipeline: PL_DLT_SCD2_Slv_To_Gld

+ Add task

Task name* Raw_to_Bronze

Type* Pipeline

Pipeline* PL_CDF_SCD1_SCD2_EMP

Trigger a full refresh on the pipeline

Depends on Select task dependencies...

Notifications + Add

Cancel Save task

Job details

Job ID 1079862481251979

Creator patel.riyas@northeastern.edu

Run as patel.riyas@northeastern.edu

Description Add description

Lineage No lineage information for this job. Learn more

Performance optimized

Schedules & Triggers

None

Add trigger

Job parameters

No job parameters are defined for this job

Edit parameters

PL_JOB - Databricks

Code modification review

Workshop Replication_CDC an: X

Untitled document - Google D: X

dbc-6043e27f-1bee.cloud.databricks.com/jobs/1079862481251979/tasks/Silver_to_Gold?o=4201568283894456

Search data, notebooks, recents, and more... CTRL + P

workspace

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

Jobs & Pipelines >

PL_JOB ☆ Lakeflow Jobs UI: ON Send feedback

Runs Tasks

Raw_to_Bronze

Pipeline: PL_CDF_SCD1_SCD2_EMP

Silver_to_Gold

Pipeline: PL_DLT_SCD2_Slv_To_Gld

+ Add task

Task name* Silver_to_Gold

Type* Pipeline

Pipeline* PL_DLT_SCD2_Slv_To_Gld

Trigger a full refresh on the pipeline

Depends on Raw_to_Bronze X

Run if dependencies All succeeded

Cancel Save task

Job details

Job ID 1079862481251979

Creator patel.riyas@northeastern.edu

Run as patel.riyas@northeastern.edu

Description Add description

Lineage No lineage information for this job. Learn more

Performance optimized

Schedules & Triggers

None

Add trigger

Job parameters

No job parameters are defined for this job

Edit parameters